Differential Equations and Slope Fields

Section 1: Slope Fields & Solution Curves

1 In the slope field for $\frac{dy}{dx} = x - y$, what is the slope at the point $(3, 1)$?

a. 2

b. 4

c. -2

d. 3

2 Along which line in the slope field for $\frac{dy}{dx} = 2x + y$ are all slope segments horizontal?

a. $y = -2x$

b. $y = 2x$

c. $x = 0$

d. $y = 0$

3 If a slope field has parallel slopes along every vertical line ($x = c$), which type of differential equation could generate it?

a. $\frac{dy}{dx} = f(x)$

b. $\frac{dy}{dx} = f(y)$

c. $\frac{dy}{dx} = x + y$

d. $\frac{dy}{dx} = xy$

Section 2: Separation of Variables

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4 Solve the differential equation $\frac{dy}{dx} = 3x^2y$ with initial condition $y(0) = 5$.

a. $y = 5e^{x^3}$

b. $y = e^{x^3} + 4$

c. $y = 5e^{3x^2}$

d. $y = x^3 + 5$

5 Solve $\frac{dy}{dx} = \frac{x}{y}$ with initial condition $y(0) = -3$.

a. $y = -\sqrt{x^2 + 9}$

b. $y = \sqrt{x^2 + 9}$

c. $y = x^2 - 3$

d. $y = -\sqrt{x^2 - 9}$

6 What is the domain of the particular solution $y = \frac{1}{2 - x}$ with $y(0) = \frac{1}{2}$?

a. $(-\infty, 2)$

b. $(2, \infty)$

c. $x \neq 2$

d. All real numbers

Section 3: Exponential Growth & Decay Models

7 Which differential equation models a quantity whose rate of growth is directly proportional to its current size?

a. $\frac{dy}{dt} = ky$

b. $\frac{dy}{dt} = k$

c. $\frac{dy}{dt} = kt$

d. $\frac{dy}{dt} = k(M - y)$

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8 A population grows according to $\frac{dP}{dt} = 0.05P$. If initial population is $1,000$, what is the population at $t = 20$?

a. $1000e$

b. $1000e^2$

c. 2000

d. $1000(1.05)^{20}$

9 A radioactive substance has a half-life of 10 years. What is its decay constant k in $\frac{dy}{dt} = -ky$?

a. $k = \frac{\ln 2}{10}$

b. $k = 10 \ln 2$

c. $k = \frac{1}{10}$

d. $k = 0.5$

Section 4: Newton's Law of Cooling

10 Newton's Law of Cooling states that $\frac{dT}{dt} = -k(T - T_m)$. What does T_m represent?

a. The ambient temperature of the surrounding medium

b. The initial temperature of the object

c. The maximum temperature reached

d. The rate of heat transfer

11 A cup of coffee at 90°C cools in a room of 20°C. If its temperature after 10 minutes is 60°C, what is the temperature difference $T - 20$ at $t = 10$?

a. 40°C

b. 70°C

c. 30°C

d. 20°C

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12 What is $\lim_{t \to \infty} T(t)$ in Newton's Law of Cooling?

a. T_m

b. 0

c. ∞

d. T_0

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